Phase 3 Strojové učenie

Importy

In [13]:
```python
import numpy as np
import pandas as pd

from sklearn.metrics import accuracy_score, precision_score, recall_score, classification_report
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.linear_model import LogisticRegression
import matplotlib.pyplot as plt

# Cesty k súborom – prispôsob ak máš iné
TRAIN_PATH = "processed/train_std.csv"
TEST_PATH = "processed/test_std.csv"
TARGET_COL = "oximetry"

train_df = pd.read_csv(TRAIN_PATH)
test_df = pd.read_csv(TEST_PATH)

train_df.head()
```

Out[13]:

	SpO₂	HR	PI	RR	EtCO₂	FiO₂	PRV	BP	Skin Temperature	Motion/Activity index	...	CO	Blood Flow Index	PPG waveform features	Signal Quality Index	Respiratory effort	O₂ extraction ratio	SNR	latitude	longitude	oximetry
0	-0.212181	-0.283977	-0.961673	-1.096729	0.671882	-1.726660	0.907547	0.747585	1.711609	-0.184376	...	-0.399321	1.737000	0.742342	1.120023	-0.563050	1.303860	0.014961	0.492058	-1.172763	1.0
1	-1.221620	1.409044	-0.531067	-0.686905	0.556444	0.497449	0.813390	0.300029	1.216281	0.196289	...	1.149775	0.628474	-0.397799	-0.089499	1.889279	-1.357021	-0.884478	0.603620	-1.284282	1.0
2	-0.018230	1.834341	1.024363	-0.509595	-0.948750	0.015322	0.789674	-0.008516	-1.216643	0.553323	...	2.145306	-0.552293	-0.141816	1.458561	-0.127506	-1.005631	0.562910	0.162366	-0.237312	0.0
3	-0.784025	1.357692	0.485828	-0.789177	-0.508072	-0.430137	0.257498	-0.242366	-1.148415	-0.365390	...	1.035635	-1.035744	0.142962	0.273825	-0.738886	-0.419158	-0.274548	0.185725	-1.765463	0.0
4	0.567762	0.743549	-2.209355	1.530722	1.175073	-0.129554	-0.927751	0.308699	1.133135	-1.391575	...	0.219173	0.477632	-1.178020	0.856171	-0.194711	-1.622471	-1.084355	0.839309	0.000765	1.0

5 rows × 23 columns

3.1 Jednoduchý klasifikátor na základe závislosti v dátach

3.1-A Naimplementujte jednoduchý ID3 klasifikátor s hĺbkou min 2 (vrátane root/koreň)

In [14]:
```python
# predpokladáme, že všetky ostatné stĺpce sú feature a oximetry je target
feature_cols = [c for c in train_df.columns if c != TARGET_COL]

# Rozdelenie na vstupne premenné a cieľovú premennú
X_train = train_df[feature_cols].values
y_train = train_df[TARGET_COL].values

X_test = test_df[feature_cols].values
y_test = test_df[TARGET_COL].values

X_train.shape, X_test.shape
```

Out[14]:
```
((9734, 22), (2434, 22))
```

Funkcie na entropiu a information gain

In [15]:
```python
def entropy(y: np.ndarray) -> float:
    """Vypočíta entropiu pre vektor tried y."""
    if len(y) == 0:
        return 0.0
    _, counts = np.unique(y, return_counts=True)
    p = counts / counts.sum()
    return -(p * np.log2(p + 1e-9)).sum()  # + malá hodnota kvôli log(0)


def best_split_for_feature(X_column: np.ndarray, y: np.ndarray):
    """
    Nájde najlepší prah (threshold) pre jeden numerický atribút
    na základe information gain.
    Vracia (najlepší_threshold, maximálny_gain). Ak sa nenašlo zlepšenie, vráti (None, 0).
    """
    # zoradíme unikátne hodnoty
    values = np.unique(X_column)
    if len(values) == 1:
        return None, 0.0

    # premenné na sledovanie najlepšieho splitu
    best_gain = 0.0
    best_threshold = None

    base_entropy = entropy(y)
    n = len(y)

    # kandidátne thresholdy – stredy medzi susednými hodnotami
    thresholds = (values[:-1] + values[1:]) / 2.0

    for t in thresholds:
        left_mask = X_column <= t
        right_mask = X_column > t

        y_left = y[left_mask]
        y_right = y[right_mask]

        if len(y_left) == 0 or len(y_right) == 0:
            continue

        ent_left = entropy(y_left)
        ent_right = entropy(y_right)

        weighted_ent = (len(y_left) / n) * ent_left + (len(y_right) / n) * ent_right
        gain = base_entropy - weighted_ent

        if gain > best_gain:
            best_gain = gain
            best_threshold = t

    return best_threshold, best_gain
```

Uzol stromu

In [16]:
```python
class TreeNode:
    def __init__(self,
                 feature_index=None,
                 threshold=None,
                 left=None,
                 right=None,
                 value=None,
                 depth=0):
        # index feature, podľa ktorého sa delí
        self.feature_index = feature_index
        # prah pre numerický split
        self.threshold = threshold
        # ľavý a pravý podstrom
        self.left = left
        self.right = right
        # hodnota listu (ak je uzol list)
        self.value = value
        # hĺbka uzla – len pre info/overfitting
        self.depth = depth

    def is_leaf(self):
        return self.value is not None
```

Implementacia ID3 klasifikátora

In [17]:
```python
class ID3Classifier:
    def __init__(self, max_depth=5, min_samples_split=2):

        #max_depth: maximálna hĺbka stromu (root má depth=0)
        #min_samples_split: minimálny počet vzoriek na ďalšie delenie

        self.max_depth = max_depth
        self.min_samples_split = min_samples_split
        self.root = None

    def _build_tree(self, X, y, depth=0):
        num_samples, num_features = X.shape
        num_labels = len(np.unique(y))

        # Stopping conditions
        # 1) všetky vzorky majú rovnaký label
        if num_labels == 1:
            leaf_value = y[0]
            return TreeNode(value=leaf_value, depth=depth)

        # 2) dosiahli sme max_depth alebo málo vzoriek
        if depth >= self.max_depth or num_samples < self.min_samples_split:
            # list – väčšinová trieda
            values, counts = np.unique(y, return_counts=True)
            leaf_value = values[np.argmax(counts)]
            return TreeNode(value=leaf_value, depth=depth)

        # Hľadanie najlepšieho rozdelenia
        best_gain = 0.0
        best_feature = None
        best_threshold = None

        for feature_idx in range(num_features):
            threshold, gain = best_split_for_feature(X[:, feature_idx], y)
            if gain > best_gain:
                best_gain = gain
                best_feature = feature_idx
                best_threshold = threshold

        # Ak žiadne zlepšenie, vytvor list
        if best_gain == 0 or best_feature is None:
            values, counts = np.unique(y, return_counts=True)
            leaf_value = values[np.argmax(counts)]
            return TreeNode(value=leaf_value, depth=depth)

        # rozdelenie dát podľa najlepšieho splitu
        left_mask = X[:, best_feature] <= best_threshold
        right_mask = X[:, best_feature] > best_threshold

        left_child = self._build_tree(X[left_mask], y[left_mask], depth + 1)
        right_child = self._build_tree(X[right_mask], y[right_mask], depth + 1)

        return TreeNode(feature_index=best_feature,
                        threshold=best_threshold,
                        left=left_child,
                        right=right_child,
                        depth=depth)

    def fit(self, X, y):
        self.root = self._build_tree(X, y, depth=0)

    def _predict_one(self, x, node: TreeNode):
        if node.is_leaf():
            return node.value

        if x[node.feature_index] <= node.threshold:
            return self._predict_one(x, node.left)
        else:
            return self._predict_one(x, node.right)

    def predict(self, X):
        return np.array([self._predict_one(x, self.root) for x in X])
```

3.1-B Vyhodnoťte Váš ID3 klasifikátor pomocou metrík accuracy, precision a recall.

In [18]:
```python
id3 = ID3Classifier(max_depth=5, min_samples_split=5)
id3.fit(X_train, y_train)

y_train_pred = id3.predict(X_train)
y_test_pred = id3.predict(X_test)

def print_metrics(name, y_true, y_pred):
    print(f"=== {name} ===")
    print("Accuracy :", accuracy_score(y_true, y_pred))
    print("Precision:", precision_score(y_true, y_pred, zero_division=0))
    print("Recall   :", recall_score(y_true, y_pred, zero_division=0))
    print()

print_metrics("ID3 – TRAIN", y_train, y_train_pred)
print_metrics("ID3 – TEST", y_test, y_test_pred)

print("Detailnejší report (test):")
print(classification_report(y_test, y_test_pred, zero_division=0))
```

```
=== ID3 – TRAIN ===
Accuracy : 0.892233408670639
Precision: 0.9116883116883117
Recall   : 0.9071330117160579

=== ID3 – TEST ===
Accuracy : 0.8935907970419064
Precision: 0.9162011173184358
Recall   : 0.9042039972432805

Detailnejší report (test):
              precision    recall  f1-score   support

         0.0       0.86      0.88      0.87       983
         1.0       0.92      0.90      0.91      1451

    accuracy                           0.89      2434
   macro avg       0.89      0.89      0.89      2434
weighted avg       0.89      0.89      0.89      2434
```

3.1-C Test na overfiting

In [19]:
```python
# "Veľmi hlboký" strom – možno overfit
id3_overfit = ID3Classifier(max_depth=10, min_samples_split=2)
id3_overfit.fit(X_train, y_train)

print_metrics("ID3 (max_depth=10) – TRAIN", y_train, id3_overfit.predict(X_train))
print_metrics("ID3 (max_depth=10) – TEST", y_test, id3_overfit.predict(X_test))

# "Regularizovaný" strom – menšia hĺbka a väčší min_samples_split
id3_regularized = ID3Classifier(max_depth=4, min_samples_split=20)
id3_regularized.fit(X_train, y_train)

print_metrics("ID3 regularized – TRAIN", y_train, id3_regularized.predict(X_train))
print_metrics("ID3 regularized – TEST", y_test, id3_regularized.predict(X_test))
```

```
=== ID3 (max_depth=10) – TRAIN ===
Accuracy : 0.9645572221080748
Precision: 0.9623920040657293
Recall   : 0.9788077188146106

=== ID3 (max_depth=10) – TEST ===
Accuracy : 0.9063270336894002
Precision: 0.9145762711864407
Recall   : 0.9297036526533425

=== ID3 regularized – TRAIN ===
Accuracy : 0.8653174440106842
Precision: 0.9252318758281279
Recall   : 0.8421778084079945

=== ID3 regularized – TEST ===
Accuracy : 0.8738701725554643
Precision: 0.9313725490196079
Recall   : 0.8511371467953136
```

Vyhodnotenie 3.1 C

Na test over fittingu sme použili 3 testovacie konfigurácie:

- **Pôvodná konfigurácia** : max depth = 5 min sample splits = 5
  - Tento model dosahoval skoro totžné výsledky na trénovacej aj testovacej sade(accuracy bolo 0.892 oproti 0.894)
  - Rozdiely boli minimálne čo nasvedčuje, že model neoverfituje a ma dostatočne dobrú generalizáciu
- **Overfit kofigurácia**: max depth = 10, min samples split = 2
  - Tu už nastával overfitting pretože si model začal *pamätať* sadu(train recal bol veľmi vysoký = 0.979)
  - Train accuracy sa zvýšilo na 0.964 a test accuracy bolo len 0.906
- **Upravená konfigurácia** : max depth = 4 min samples split = 20
  - Pri menšej maximálnej hĺbke a väčšom splite sme dosiahly train accuraycy 0.896 a test accuracy 0.874 čím rozdiel skoro úplne zmizol
  - Dôsledkom toho bolo zníženie overfittingu a model sa naučil viac všeobecnié vzory

3.2 Trénovanie a vyhodnotenie klasifikátorov strojového učenia

3.2-A Na trénovanie využite jeden stromový algoritmus v scikit-learn.

In [20]:
```python
dt_clf = DecisionTreeClassifier(
    max_depth=5,           # podobné nastavenie ako náš ID3
    min_samples_split=5,
    random_state=42
)

dt_clf.fit(X_train, y_train)

y_train_pred_dt = dt_clf.predict(X_train)
y_test_pred_dt = dt_clf.predict(X_test)

print_metrics("DecisionTree – TRAIN", y_train, y_train_pred_dt)
print_metrics("DecisionTree – TEST", y_test, y_test_pred_dt)
```

```
=== DecisionTree – TRAIN ===
Accuracy : 0.8992192315594822
Precision: 0.9262860173236698
Recall   : 0.9028256374913852

=== DecisionTree – TEST ===
Accuracy : 0.8940016433853739
Precision: 0.923951670220327
Recall   : 0.895933838731909
```

3.2-B Porovnajte s jedným iným nestromovým algoritmom v scikit-learn

In [21]:
```python
log_reg = LogisticRegression(
    max_iter=1000,        # pre istotu viac iterácií, aby konvergoval
)

log_reg.fit(X_train, y_train)

y_train_pred_lr = log_reg.predict(X_train)
y_test_pred_lr = log_reg.predict(X_test)

print_metrics("LogisticRegression – TRAIN", y_train, y_train_pred_lr)
print_metrics("LogisticRegression – TEST", y_test, y_test_pred_lr)
```

```
=== LogisticRegression – TRAIN ===
Accuracy : 0.8462091637559072
Precision: 0.8671781756180733
Recall   : 0.8762922122674018

=== LogisticRegression – TEST ===
Accuracy : 0.8578471651602301
Precision: 0.8745762711864407
Recall   : 0.8890420399724328
```

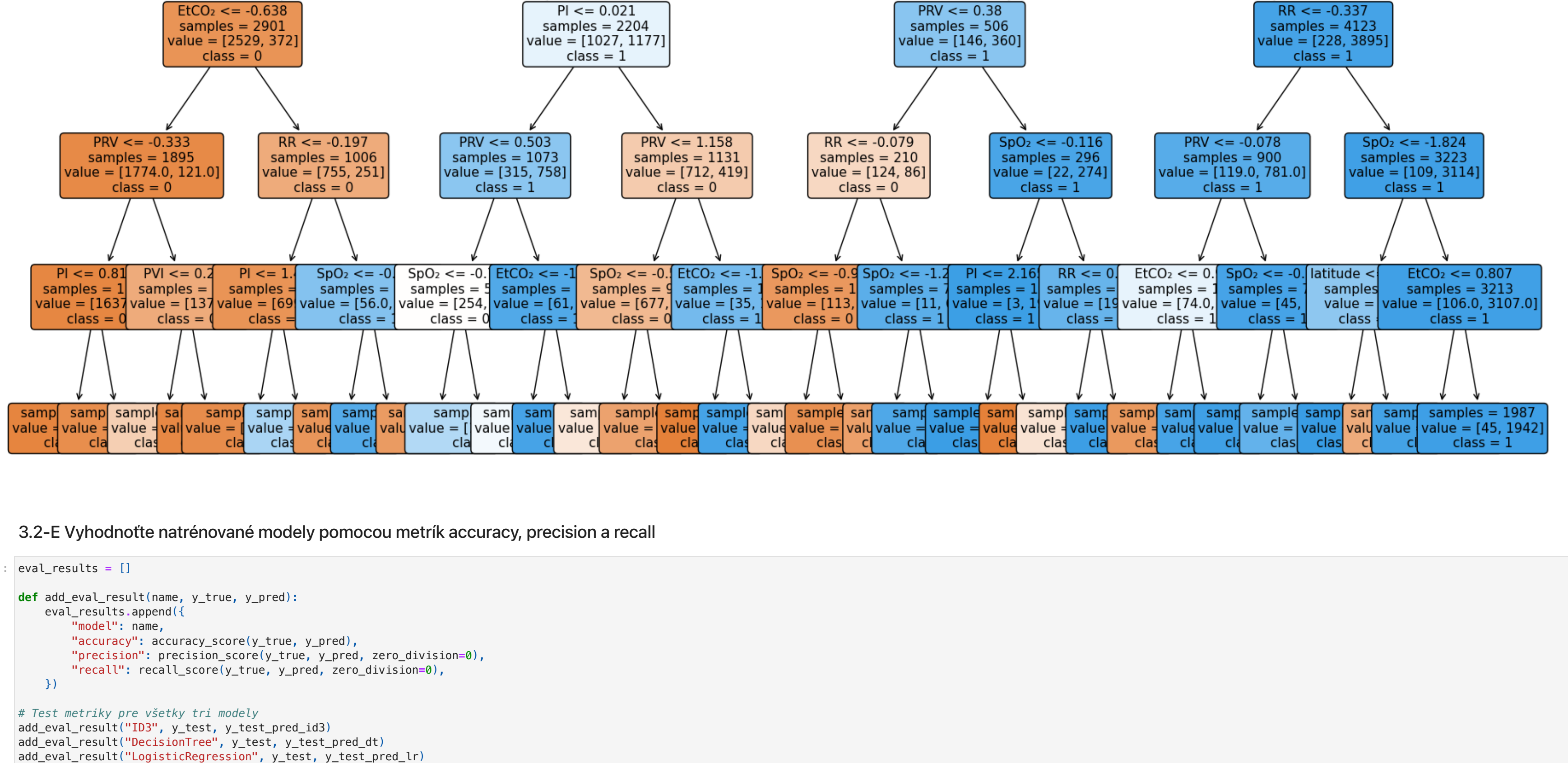
3.2-C Porovnajte výsledky s ID3 z prvého kroku.

In [22]:
```python
# Predikcie pre všetky tri modely – pre istotu ich spočítame znova

# 1. ID3
y_train_pred_id3 = id3.predict(X_train)
y_test_pred_id3 = id3.predict(X_test)

# 2. DecisionTreeClassifier (sklearn)
y_train_pred_dt = dt_clf.predict(X_train)
y_test_pred_dt = dt_clf.predict(X_test)

# 3. LogisticRegression (sklearn)
y_train_pred_lr = log_reg.predict(X_train)
y_test_pred_lr = log_reg.predict(X_test)

print("=== Porovnanie accuracy (TRAIN) ===")
print("ID3 (vlastný)       :", accuracy_score(y_train, y_train_pred_id3))
print("DecisionTree (skl.) :", accuracy_score(y_train, y_train_pred_dt))
print("LogisticReg (skl.)  :", accuracy_score(y_train, y_train_pred_lr))

print("\n=== Porovnanie accuracy (TEST) ===")
print("ID3 (vlastný)       :", accuracy_score(y_test, y_test_pred_id3))
print("DecisionTree (skl.) :", accuracy_score(y_test, y_test_pred_dt))
print("LogisticReg (skl.)  :", accuracy_score(y_test, y_test_pred_lr))
```

```
=== Porovnanie accuracy (TRAIN) ===
ID3 (vlastný)       : 0.892233408670639
DecisionTree (skl.) : 0.8992192315594822
LogisticReg (skl.)  : 0.8462091637559072

=== Porovnanie accuracy (TEST) ===
ID3 (vlastný)       : 0.8935907970419064
DecisionTree (skl.) : 0.8940016433853739
LogisticReg (skl.)  : 0.8578471651602301
```

3.2-D Vizualizujte natrénované pravidlá minimálne pre jeden Vami vybraný algoritmus

In [23]:
```python
plt.figure(figsize=(22, 12))
plot_tree(
    dt_clf,
    feature_names=feature_cols,
    class_names=["0", "1"],
    filled=True,
    max_depth=5,
    impurity=False,
    proportion=False,
    rounded=True,
    fontsize=11
)
plt.show()
```



3.2-E Vyhodnoťte natrénované modely pomocou metrík accuracy, precision a recall

In [24]:
```python
eval_results = []

def add_eval_result(name, y_true, y_pred):
    eval_results.append({
        "model": name,
        "accuracy": accuracy_score(y_true, y_pred),
        "precision": precision_score(y_true, y_pred, zero_division=0),
        "recall": recall_score(y_true, y_pred, zero_division=0),
    })

# Test metriky pre všetky tri modely
add_eval_result("ID3", y_test, y_test_pred_id3)
add_eval_result("DecisionTree", y_test, y_test_pred_dt)
add_eval_result("LogisticRegression", y_test, y_test_pred_lr)

eval_df = pd.DataFrame(eval_results)
eval_df
```

Out[24]:

	model	accuracy	precision	recall
0	ID3	0.893591	0.916201	0.904204
1	DecisionTree	0.894002	0.923952	0.895934
2	LogisticRegression	0.857847	0.874576	0.889042

3.3 Optimalizácia alias hyperparameter tuning

3.3 (A) Na trénovanie využite jeden stromový algoritmus v scikit-learn